

Latent Variable Models and E-M algorithm

Statistical Foundations of AI and ML, Module 4

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Generative Model for Clustering

- ▶ Generative model is a story about how the data was created
- ▶ We imagine that each of K clusters has a prototype
- ▶ Every data point is a “noisy version” of one prototype
- ▶ For any datapoint i ,
 - ▶ first the cluster index Z_i is decided ($Z_i \in \{1, 2, \dots, K\}$)
 - ▶ then the feature X_i is created, as a noisy version of the selected cluster's prototype

Generative Model for Clustering

- ▶ Assumption: Each cluster represented by prototypes: $\{\theta_1, \theta_2, \dots, \theta_K\}$
- ▶ for each datapoint i
 - ▶ Draw cluster index $Z_i \sim g$ (g : distribution on the clusters)
 - ▶ Draw feature vector $X_i \sim f(\theta_{Z_i})$ (f : distribution on the observation space)
- ▶ We choose f and g according to application (eg. f can be Gaussian if our observations are real-valued)

Inference and Estimation Problems

- ▶ Observed variables: X (our observed datapoints)
- ▶ Unknown variables: cluster assignments Z , cluster parameters θ
- ▶ Finding Z : Inference problem, $prob(Z|X, \theta)$
- ▶ Finding θ : Estimation problem, $\theta = argmaxprob(Z, X, \theta)$
- ▶ Challenge: The two problems are linked together!
- ▶ Cannot estimate θ directly because of Z

Gaussian Mixture Model

- ▶ Each cluster represented by a Gaussian distribution: $\mathcal{N}(\mu_j, \sigma_j)$ ($j \in \{1, \dots, K\}$)
- ▶ Each cluster has a probability π_j , $\pi = [\pi_1, \dots, \pi_K]$, $\pi_j \geq 0$, $\sum_{j=1}^K \pi_j = 1$
- ▶ Model parameters: $\{\mu_j, \sigma_j, \pi_j\}_{j=1}^K$
- ▶ for each datapoint i
 - ▶ Draw cluster index $Z_i \sim \text{Categorical}(\pi)$
 - ▶ Draw feature vector $X_i \sim \mathcal{N}(\mu_{Z_i}, \sigma_{Z_i})$

Gaussian Mixture Model

- ▶ X_i depends only on Z_i , Z_i depends on nothing!
- ▶ Joint distribution:
$$\text{prob}(Z_1, \dots, Z_N, X_1, \dots, X_N) = \prod_{i=1}^N \text{prob}(Z_i) \text{prob}(X_i|Z_i)$$
- ▶ $\text{prob}(Z_i) = \prod_{j=1}^K \pi_j^{I(Z_i=j)}$ (I : indicator function)
- ▶ $\text{prob}(X_i|Z_i) = \prod_{j=1}^K \left(\frac{1}{\sigma_j} \exp\left(-\frac{(x_i - \mu_j)^2}{2\sigma_j^2}\right) \right)^{I(Z_i=j)}$
- ▶ Likelihood function
$$\mathcal{L}(\mu, \sigma, \pi) = \prod_{i=1}^N \prod_{j=1}^K \left(\frac{\pi_j}{\sigma_j} \exp\left(-\frac{(x_i - \mu_j)^2}{2\sigma_j^2}\right) \right)^{I(Z_i=j)}$$
- ▶ Log-likelihood =
$$\sum_{i=1}^N \sum_{j=1}^K I(Z_i = j) \left(\log \pi_j - \log \sigma_j - \frac{(x_i - \mu_j)^2}{2\sigma_j^2} \right)$$

Gaussian Mixture Model

- ▶ $\{\mu_{MLE}, \sigma_{MLE}, \pi_{MLE}\} = \operatorname{argmax}_{\mu, \sigma, \pi} \mathcal{L}(\mu, \sigma, \pi)$
- ▶ Solve $\frac{\partial \mathcal{L}}{\partial \mu_j} = 0, \frac{\partial \mathcal{L}}{\partial \sigma_j} = 0$
- ▶ $\mu_j = \frac{\sum_{i=1}^N I(Z_i=j) x_i}{\sum_{i=1}^N I(Z_i=j)}$, i.e. mean of the points in cluster j
- ▶ $\sigma_j = \frac{\sum_{i=1}^N I(Z_i=j) (x_i - \mu_j)^2}{\sum_{i=1}^N I(Z_i=j)}$ i.e. variance of the points in cluster j
- ▶ $\pi_j = \frac{\sum_{i=1}^N I(Z_i=j)}{\sum_{i=1}^N \sum_{j=1}^N I(Z_i=j)}$, i.e. relative frequency of the points in cluster j
- ▶ Unfortunately we cannot compute these, as we do not know Z !

Expectation Maximization

- ▶ As we do not know $I(Z_i = j)$, we consider it as a random variable, with distribution $p(Z_i|X)$
- ▶ We replace $I(Z_i = j)$ by its expected value, $\gamma_{ij} = E(I(Z_i = j))$
- ▶ As I is binary, $E(I(Z_i = j)) = p(Z_i = j|X)$
- ▶
$$p(Z_i = j|X) = p(Z_i = j|X_i) = \frac{p(X_i|Z_i=j)p(Z_i=j)}{p(X_i)} = \frac{p(X_i|Z_i=j)p(Z_i=j)}{\sum_{l=1}^K p(X_i|Z_i=l)p(Z_i=l)}$$
- ▶ So,
$$\gamma_{ij} = \frac{\pi_j \mathcal{N}(X_i; \mu_j, \sigma_j)}{\sum_{l=1}^K \pi_l \mathcal{N}(x_i; \mu_l, \sigma_l)}$$
- ▶
$$\mu_j = \frac{\sum_{i=1}^N \gamma_{ij} x_i}{\sum_{i=1}^N \gamma_{ij}}, \sigma_j = \frac{\sum_{i=1}^N \gamma_{ij} (x_i - \mu_j)^2}{\sum_{i=1}^N \gamma_{ij}}, \pi_j = \frac{\sum_{i=1}^N \gamma_{ij}}{\sum_{i=1}^N \sum_{j=1}^K \gamma_{ij}}$$

Expectation Maximization

► We use an iterative algorithm

1. Initialize μ^0, σ^0, π^0

2. Repeat

2.1 E-step: Calculate $\gamma_{ij} = \frac{\pi_j^0 \mathcal{N}(X_i; \mu_j^0, \sigma_j^0)}{\sum_{l=1}^K \pi_l^0 \mathcal{N}(X_i; \mu_l^0, \sigma_l^0)}$

2.2 M-step: Re-estimate the parameters

2.3 $\mu_j^1 = \frac{\sum_{i=1}^N \gamma_{ij} x_i}{\sum_{i=1}^N \gamma_{ij}}, \sigma_j^1 = \frac{\sum_{i=1}^N \gamma_{ij} (x_i - \mu_j^1)^2}{\sum_{i=1}^N \gamma_{ij}}, \pi_j^1 = \frac{\sum_{i=1}^N \gamma_{ij}}{N}$

3. If $(\mu^0, \sigma^0, \pi^0) \approx (\mu^1, \sigma^1, \pi^1)$, STOP

4. Else set $(\mu^0 = \mu^1, \sigma^0 = \sigma^1, \pi^0 = \pi^1)$ and GOTO 2

Expectation Maximization

- ▶ When E-M algorithm converges, we get optimal values of the parameters $(\mu^{EM}, \sigma^{EM}, \pi^{EM})$
- ▶ Compute posterior distribution $p(Z_i|X_i) = \frac{\pi_j^{EM} \mathcal{N}(X_i; \mu_j^{EM}, \sigma_j^{EM})}{\sum_{l=1}^K \pi_l \mathcal{N}(X_i; \mu_l^{EM}, \sigma_l^{EM})}$
- ▶ Soft-clustering instead of hard-clustering as in K-means
- ▶ Mode of distribution may be used as cluster assignment

Model Likelihood

- ▶ The likelihood of a model: $\mathcal{L}(P) = \text{prob}(X)$ the joint distribution of the data according to the model
- ▶ If model contains *latent variables* like Z , marginalize over them

$$\begin{aligned}\mathcal{L}(\mu, \sigma, \pi) = \text{prob}(X) &= \prod_{i=1}^N \text{prob}(X_i) = \prod_{i=1}^N \sum_{k=1}^K \text{prob}(X_i, Z_i = k) \\ &= \prod_{i=1}^N \sum_{k=1}^K \text{prob}(X_i | Z_i = k) \text{prob}(Z_i = k) \\ &= \prod_{i=1}^N \sum_{k=1}^K \pi_k \frac{1}{2\pi\sigma_k} \exp\left(-\frac{(x_i - \mu_k)^2}{2\sigma_k^2}\right)\end{aligned}$$

Comparing models

- ▶ Two different GMMs - with different sets of parameters (μ_a, σ_a, π_a) and (μ_b, σ_b, π_b)
- ▶ They can be compared by their likelihoods
- ▶ $\mathcal{L}(\mu_a, \sigma_a, \pi_a) > \mathcal{L}(\mu_b, \sigma_b, \pi_b)$ implies that first model *fits* the data better than the second
- ▶ Choosing K may be done by this approach

Latent Linear Gaussian Model

- ▶ $Z_i \sim \mathcal{N}(\mu_0, \Sigma_0)$, $X_i \sim \mathcal{N}(WZ_i + \mu, \Sigma)$
- ▶ Observation $X = \{X_1, \dots, X_N\}$, Parameters $\theta = \{W, \mu_0, \mu, \Sigma_0, \Sigma\}$, Latent $Z = \{Z_1, \dots, Z_N\}$
- ▶ Posterior on latents $p_\theta(Z_i|X_i) = \mathcal{N}(\mu_i, \Sigma_i)$ where $\Sigma_i = (\Sigma_0^{-1} + W^T \Sigma W)^{-1}$ and $\mu_i = \Sigma_i(W^T \Sigma^{-1}(X_i - \mu) + \Sigma_0^{-1} \mu_0)$
- ▶ Marginal $p_\theta(X_i) = \mathcal{N}(W\mu_0 + \mu, \Sigma + W\Sigma_0 W^T)$
- ▶ Log-Likelihood $l(\theta) = \sum_{i=1}^N \log(p_\theta(X_i))$
- ▶ Parameters can be estimated by maximum-likelihood

Mixture of Latent Linear Gaussians

- ▶ $Y_i \sim \text{Cat}(\pi)$, $Z_i \sim \mathcal{N}(\mu_0, \Sigma_0)$
- ▶ $X_i | Z_i, Y_i = k \sim \mathcal{N}(W_k Z_i + \mu_k, \Sigma_k)$
- ▶ Observation $X = \{X_1, \dots, X_N\}$, Latent $Z = \{Z_1, \dots, Z_N, Y_1, \dots, Y_N\}$
- ▶ Parameters
 $\theta = \{\pi, W_1, \dots, W_K, \mu_0, \mu_1, \dots, \mu_K, \Sigma_0, \Sigma_1, \dots, \Sigma_K\}$
- ▶ For simplicity, assume $\mu_0 = 0, \Sigma_0 = I, \Sigma_k = \Sigma$
- ▶ Log-likelihood $l(\theta) = \sum_{i=1}^N \log(p_\theta(X_i))$
- ▶ Expected Complete Log-likelihood $E_{r,s}(\log(p_\theta((X, Y, Z)))$
where $r_i(Y_i) = p_\theta(Y_i | X_i)$ and $s_i(Z_i) = p_\theta(Z_i | X_i, Y_i)$

E-M for Mixture of Latent Linear Gaussians

- ▶ E-step for Y :

$$r_i(Y_i = c, \theta^t) = p_{\theta^t}(Y_i = c | X_i) \propto \pi_c^t \mathcal{N}(\mu_c^t, W_c^t W_c^{tT})$$

- ▶ E-step for Z :

$$s_i(Z_i | X_i, Y_i, \theta^t) = p_{\theta^t}(Z_i | X_i, Y_i = c) = \mathcal{N}(m_{ic}, \Sigma_{ic}) \text{ where } \Sigma_{ic} = (I + W_c^{tT} \Sigma^t W_c^t)^{-1} \text{ and } m_{ic} = \Sigma_{ic}(W_c^{tT} \Sigma^{-1}(X_i - \mu_c))$$

- ▶ M-step: Estimate θ^{t+1} as below ($\hat{W}_c = \{W_c, \mu_c\}$, $q_i = Y_i$):

(W_c, μ_c) , $\bar{z} = (z, 1)$, Also, define

$$\mathbf{b}_{ic} \triangleq \mathbb{E}[\bar{\mathbf{z}} | \mathbf{x}_i, q_i = c] = [\mathbf{m}_{ic}; 1]$$

$$\mathbf{C}_{ic} \triangleq \mathbb{E}[\bar{\mathbf{z}} \bar{\mathbf{z}}^T | \mathbf{x}_i, q_i = c] = \begin{pmatrix} \mathbb{E}[\mathbf{z} \mathbf{z}^T | \mathbf{x}_i, q_i = c] & \mathbb{E}[\mathbf{z} | \mathbf{x}_i, q_i = c] \\ \mathbb{E}[\mathbf{z} | \mathbf{x}_i, q_i = c]^T & 1 \end{pmatrix}$$

Then the M step is as follows:

$$\begin{aligned} \hat{W}_c &= \left[\sum_i r_{ic} \mathbf{x}_i \mathbf{b}_{ic}^T \right] \left[\sum_i r_{ic} \mathbf{C}_{ic} \right]^{-1} \\ \hat{\Psi} &= \frac{1}{N} \text{diag} \left\{ \sum_{ic} r_{ic} (\mathbf{x}_i - \hat{W}_c \mathbf{b}_{ic}) \mathbf{x}_i^T \right\} \\ \hat{\pi}_c &= \frac{1}{N} \sum_{i=1}^N r_{ic} \end{aligned}$$

Hidden Markov Model

- ▶ Consider sequential observations $x_1, x_2, \dots, x_T$
- ▶ Key assumption in GMM: all the data-points are independent
- ▶ For sequential applications, this may not be true any longer!
- ▶ eg. a long audio stream with many speakers
- ▶ The observation x_t is likely to belong to same speaker as x_{t-1}
- ▶ There may be a transition pattern from one speaker to another!

Hidden Markov Model

- ▶ Different values of Z indicate *state* of the system (eg. which speaker is talking)
- ▶ System may have K states (decided by user)
- ▶ Current state Z_t depends on previous states $Z_1, \dots, Z_{t-1}$
- ▶ Instead of $prob(Z_t)$, we need $prob(Z_t|Z_{t-1}, \dots, Z_1)$
- ▶ *Markov Assumption*: Future independent of past, given the present!
- ▶ Markov model: $prob(Z_t|Z_{t-1}, \dots, Z_1) = prob(Z_t|Z_{t-1})$
- ▶ New parameter instead of π : $A_{ij} = prob(Z_t = j|Z_{t-1} = i)$

Hidden Markov Model

- ▶ Each state represented by parameters: p_j ($j \in \{1, \dots, K\}$) of *emission distribution* f
- ▶ *Transition distribution* from state i to state j :
 $A_{ij} = \text{prob}(Z_t = j | Z_{t-1} = i)$ ($K \times K$ matrix)
- ▶ Each row of matrix A : categorical probability distribution
- ▶ An *initial state distribution* π (similar to GMM)
- ▶ $Z_1 \sim \text{Categorical}(\pi)$; $X_1 \sim f(p_{Z_1})$
- ▶ for each datapoint t
 - ▶ Draw cluster index $Z_t \sim \text{Categorical}(A_{Z_{t-1}})$
 - ▶ Draw feature vector $X_t \sim f(p_{Z_t})$

Hidden Markov Model

- ▶ Common emission distributions: Categorical (discrete observations) or Gaussian (real observations)
- ▶ X_t depends on Z_t only, Z_t depends on Z_{t-1} only
- ▶ Joint distribution $prob(X, Z) = prob(Z_1)prob(X_1|Z_1) \prod_{t=2}^T prob(Z_t|Z_{t-1})prob(X_t|Z_t)$
- ▶ Rearranging, $prob(X, Z) = prob(Z_1) \times \prod_{t=2}^T prob(Z_t|Z_{t-1}) \times \prod_{t=1}^T prob(\cancel{Z_t} | \cancel{Z_{t-1}}) \times \prod_{t=1}^T prob(X_t | Z_t)$
- ▶ $prob(Z_1) : \pi$ (initial state distribution), $prob(Z_t|Z_{t-1}) : A$ (transition distribution), $prob(X_t|Z_t) : f(p)$ (emission distribution)

Forward-Backward Algorithm

Inference problem: Given (π, A, p) , find posterior distribution $prob(Z_t|X_1, \dots, X_T)$

$$\begin{aligned} prob(Z_t|X_1, \dots, X_T) &\propto prob(Z_t, X_1, \dots, X_T) \\ &= prob(Z_t, X_1, \dots, X_t) \\ &\times prob(X_{t+1}, \dots, X_T|Z_t, X_1, \dots, X_t) \\ &= prob(Z_t, X_1, \dots, X_t) prob(X_{t+1}, \dots, X_T|Z_t) \\ &= \alpha_t(Z_t) \beta_t(Z_t) \end{aligned}$$

Forward Algorithm

$$\begin{aligned}\alpha_t(Z_t) &= \text{prob}(Z_t, X_1, \dots, X_t) \\&= \sum_{Z_{t-1}} \text{prob}(Z_t, Z_{t-1}, X_1, \dots, X_t) \\&= \sum_{Z_{t-1}} \text{prob}(Z_{t-1}, X_1, \dots, X_{t-1}) \text{prob}(Z_t, X_t | Z_{t-1}, X_1, \dots, X_{t-1}) \\&= \sum_{Z_{t-1}} \alpha_{t-1}(Z_{t-1}) \text{prob}(Z_t, X_t | Z_{t-1}) \\&= \sum_{Z_{t-1}} \alpha_{t-1}(Z_{t-1}) \text{prob}(Z_t | Z_{t-1}) \text{prob}(X_t | Z_t)\end{aligned}$$

$$\alpha_1(Z_1) = \prod_{k=1}^K \pi_k^{I(Z_1=k)} f(X_1, p_k), \text{prob}(Z_t | Z_{t-1}) = \prod_{k,l=1}^{K,K} A_{kl}^{I(Z_{t-1}=k, Z_t=l)}$$

Backward Algorithm

$$\begin{aligned}\beta_t(Z_t) &= \text{prob}(X_{t+1}, \dots, X_T | Z_t) \\ &= \sum_{Z_{t+1}} \text{prob}(Z_{t+1}, X_{t+1}, \dots, X_T | Z_t) \\ &= \sum_{Z_{t+1}} \text{prob}(Z_{t+1}, X_{t+1} | Z_t) \text{prob}(X_{t+2}, \dots, X_T | Z_{t+1}, X_{t+1}, Z_t) \\ &= \sum_{Z_{t+1}} \text{prob}(Z_{t+1} | Z_t) \text{prob}(X_{t+1} | Z_{t+1}) \text{prob}(X_{t+2}, \dots, X_T | Z_{t+1}) \\ &= \sum_{Z_{t+1}} \text{prob}(Z_{t+1} | Z_t) \text{prob}(X_{t+1} | Z_{t+1}) \beta_{t+1}(Z_{t+1})\end{aligned}$$

$$\begin{aligned}\beta_{T-1}(Z_{T-1}) &= \sum_{Z_T} \text{prob}(X_T, Z_T | Z_{T-1}) \\ &= \sum_{Z_T} \text{prob}(Z_T | Z_{T-1}) \text{prob}(X_T | Z_T)\end{aligned}$$

Parameter Estimation in HMM

Estimation problem: Estimate the parameters (π, A, p) , though we don't know Z

$$\text{prob}(X, Z) = \text{prob}(Z_1) \text{prob}(X_1|Z_1) \prod_{t=2}^T \text{prob}(Z_t|Z_{t-1}) \text{prob}(X_t|Z_t)$$

$$\mathcal{L}(\pi, A, p) = \prod_{k=1}^K \pi_k^{I(Z_1=k)} \times \prod_{t=2}^T \prod_{k,l=1}^{K,K} A_{kl}^{I(Z_{t-1}=k, Z_t=l)} \times \prod_{t=1}^T f(X_t, p_{Z_t})$$

Replace $I(Z_1 = k)$ by $\gamma_1(k) = E(I(Z_1 = k))$, $I(Z_{t-1} = k, Z_t = l)$ by $\xi_t(kl) = E(I(Z_{t-1} = k, Z_t = l))$

Baum-Welch Algorithm (E-M)

Input: sequence $\{X_1, \dots, X_T\}$, emission parameters p

1. Make initial estimates of parameters π^0, A^0

2. Repeat

$$2.1 \quad \pi_k^1 = \gamma_k = \frac{\pi_k^0 f(X_1, p_k)}{\sum_{l=1}^K \pi_l^0 f(X_1, p_l)}$$

$$2.2 \quad A_{kl}^1 = \frac{\sum_{t=1}^{T-1} \xi_t(kl)}{\sum_{t=1}^{T-1} \gamma_t(k)}$$

2.3 If $(\pi^0, A^0) \approx (\pi^1, A^1)$, STOP

2.4 Else set $(\pi^0 = \pi^1, A^0 = A^1)$ and GOTO 2

$$\gamma_t(k) = \frac{\alpha_t(k) \beta_t(k)}{\sum_{i=1}^K \alpha_t(i) \beta_t(i)}, \quad \xi_t(kl) = \frac{\alpha_{t-1}(k) \beta_t(l) A_{kl}^0 f(X_t, p_l)}{\sum_{i,j=1}^{K,K} \alpha_{t-1}(i) \beta_t(j) A_{ij}^0 f(X_t, p_j)}$$

Why does E-M work?

- ▶ Latent Variables: Z , Observed variables: X , Model parameters: θ
- ▶ Full model log-likelihood $\log(p_\theta(Z, X))$ cannot be evaluated because Z is latent
- ▶ Data log-likelihood $l(\theta) = \log(p_\theta(X)) = \log(\sum_Z p_\theta(X, Z))$
- ▶ Using Jensen's Inequality, $\log(E_q(Z)) \geq E_q(\log(Z))$ for any arbitrary distribution q
- ▶ Hence $l(\theta) = \log(\sum_Z q(Z) \frac{p_\theta(X, Z)}{q(Z)}) \geq \sum_{q(Z)} \log(\frac{p_\theta(X, Z)}{q(Z)})$
- ▶ i.e. $l(\theta) \geq E_q(\log(\frac{p_\theta(X, Z)}{q(Z)})) = Q(\theta, q)$
- ▶ So for any arbitrary distribution q , $Q(\theta, q)$ is a **lower bound** on q

Why does E-M work?

- ▶ Aim: to estimate the parameters θ that maximize $l(\theta)$ - which is analytically difficult
- ▶ **Key idea:** As an alternative, find a **tight lower bound** of $l(\theta)$ and maximize it
- ▶ Find that distribution $q(Z)$ for which $Q(\theta, q)$ is as close to $l(\theta)$ as possible!
- ▶ $Q(\theta, q) = E_q(\log(\frac{p_\theta(X, Z)}{q(Z)})) = E_q(\log(\frac{p_\theta(Z/X)p_\theta(X)}{q(Z)})) = E_q(\log(\frac{p_\theta(Z/X)}{q(Z)}) + \log(p_\theta(X)))$
- ▶ Rearranging terms, we have $l(\theta) = Q(\theta, q) + KL(p_\theta(Z/X), q(Z))$
- ▶ If $q(Z) = p_\theta(Z/X)$, KL is 0, i.e. $l(\theta) = Q(\theta, q)$ (tight lower bound)

Why does E-M work?

- ▶ Now that we have found a tight lower bound, we need to estimate parameters θ to maximize it
- ▶ But since we cannot maximize it directly, we have to do it iteratively!
- ▶ Current estimate of parameters: θ^t
- ▶ $q^t(Z) = p_{\theta^t}(Z|X)$, i.e. conditional distribution of latent variables w.r.t. current estimate of parameters
- ▶ This can be numerically evaluated using the model
- ▶ $Q(\theta, q^t) = E_{q^t}(\log(\frac{p_{\theta}(X, Z)}{q^t(Z)})) = E_{q^t}(\log(p_{\theta}(X, Z))) + C$
- ▶ Calculating this is equivalent to the E-step (calculating expected total likelihood WRT current parameter estimate)!

Why does E-M work?

- ▶ $\theta^{t+1} = \operatorname{argmax}_{\theta} Q(\theta, q^t) = \operatorname{argmax}_{\theta} E_{q^t}(\log(p_{\theta}(X, Z)))$
- ▶ This is equivalent to maximizing the total likelihood
- ▶ $l(\theta^{t+1}) \geq Q(\theta^{t+1}, q^t)$ (Jensen's Inequality)
- ▶ $Q(\theta^{t+1}, q^t) \geq Q(\theta^t, q^t)$ (As θ^{t+1} maximizes $Q(\theta, q^t)$)
- ▶ But $Q(\theta^t, q^t) = l(\theta^t)$ (tight lower bound)
- ▶ Combining, we have $l(\theta^{t+1}) \geq l(\theta^t)$, i.e. with each iteration the data likelihood increases
- ▶ Hence, E-M algorithm must converge to a maxima (local/global) depending on initial values θ^0

General Framework of E-M

- ▶ Find analytical expression for complete log-likelihood of model $\log(p_{\theta}(Z, X))$
- ▶ Find analytical expression of expected complete likelihood $E_q(\log(p_{\theta}(Z, X)))$, where the expectation is taken with respect to $q(Z) = p_{\theta}(Z|X)$, i.e. posterior of latent variables WRT observations
- ▶ Initialize the parameters θ^0
- ▶ **E-step**: Calculate numerically the posterior of latent variables WRT this parameter estimates $q^0(Z) = p_{\theta^0}(Z|X)$
- ▶ **M-step**: Choose θ^1 to maximize the expected complete likelihood $E_{q^0}(\log(p_{\theta}(Z, X)))$
- ▶ Repeat E and M steps till θ converges

Gibbs Sampling

- ▶ Consider a high-dimensional distribution over $\mathcal{R}^D$
- ▶ Sampling new points more difficult in H-D space, most points likely to be rejected
- ▶ **Approximation:** move in one dimension at a time!
- ▶ Choose a dimension d , and search candidate values as $g(x_d | x_1^t, \dots, x_{d-1}^t, x_{d+1}^t, \dots, x_D^t)$
- ▶ If sampled value x_d^{new} (obtained from g) is accepted, set $x^{t+1} = \{x_1^t, \dots, x_{d-1}^t, x_d^{new}, x_{d+1}^t, \dots, x_D^t\}$
- ▶ Repeat above steps for all dimensions in order, over and over
- ▶ Eliminate Accept-Reject decision by using f itself as transition distribution g
- ▶ Feasible only if $f(x_d | x_1, \dots, x_{d-1}, x_{d+1}, \dots, x_D)$ can be calculated

Inference in Latent Variable Models

- ▶ Gibbs Sampling is a very useful tool of inference in Latent Variable Models
- ▶ Example: Estimation of Z -variables in Hidden Markov Models, $p(Z_t|X)$
- ▶ We need $p(Z_t|X, Z_{-t})$ where $Z_{-t} = \{Z_1, \dots, Z_{t-1}, Z_{t+1}, \dots, Z_T\}$
- ▶ $p(Z_t = k|X, Z_{-t}) \propto p(Z, Z_{-t}, Z_t = k) = p(Z_{t+1}|Z_t = k)p(Z_t = k|Z_{t-1})p(X_t|Z_t = k)$ (using HMM model)
- ▶ We know X and have fixed Z_{-t} for now
- ▶ $p(Z_{t+1}|Z_t = k)$ and $p(Z_t = k|Z_{t-1})$ can be easily calculated from the state transition distribution
- ▶ $p(X_t|Z_t = k)$ can be calculated from emission distribution
- ▶ Based on these, we can sample new value of Z_t from $p(Z_t|X, Z_{-t})$

Gibbs Sampling for Latent Variable Models

- ▶ General framework of Gibbs Sampling is applicable for all latent variable-based models
- ▶ Calculate conditional distributions of each latent variable $p(Z_i|Z_{-i}, X)$
- ▶ Initialize all latent variables by any suitable approach (eg. clustering)
- ▶ For each iteration
 - ▶ Sample each latent variable in order from $p(Z_i|Z_{-i}, X)$ and update it
 - ▶ Use the updated values of each latent variables for sampling the next
- ▶ If iteration is within burn-in period (say first 50 iterations), just carry on to next iteration
- ▶ Else, if iteration is at chosen interval (eg. $\text{mod}(\text{iteration}, 10) = 0$), store the current values of all latent variables

Gibbs Sampling for Latent Variable Models

- ▶ Finally, we end up with many samples from each latent variable
- ▶ We can now calculate posterior distribution $p(Z_i|X)$ or MAP estimate of each latent variable Z_i
- ▶ Burn-in samples are discarded as Markov Chain hasn't mixed yet (we are not sampling from the stationary distribution $P(Z|X)$ as yet)
- ▶ After burn-in, sample values stored at regular intervals to avoid correlation between successive draws (we want IID samples)
- ▶ Each stored sample basically gives us a snapshot of the latent variables at that point
- ▶ There is no concept of convergence (i.e. the latent variables will not become fixed beyond some iteration)
- ▶ Rather, the stored values of each random variable will reflect its posterior distribution!